

Speech Denoising in the Short-Time Fourier Transform Domain: A Comparative Study of Classical DSP Methods

Mid-Semester Evaluation Report
EE 317 — Digital Signal Processing

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1. Introduction and Motivation

1.1 Background

Every audio signal acquired in the real world is corrupted. Thermal noise in the microphone and preamplifier, quantisation error in the analog-to-digital converter, acoustic interference from the surrounding environment, and channel noise introduced during transmission all combine so that the signal reaching the receiver is never the signal that was produced at the source. Formally, the observed signal is modelled as

$$x[n] = s[n] + d[n]$$

where $s[n]$ is the clean speech, $d[n]$ is additive noise, and $x[n]$ is what we actually measure. The task of denoising is to recover an estimate $\hat{s}[n]$ from $x[n]$ alone, without access to either the clean signal or the noise in isolation.

1.2 Why this matters

Degraded audio is not merely unpleasant to listen to. It has measurable downstream consequences across a range of systems:

- **Telephony and voice over IP.** Perceived call quality drops sharply with background noise, and codecs allocate bits inefficiently when fed noisy input.

- **Automatic speech recognition.** ASR word error rates degrade severely under mismatched noise conditions. Denoising is a standard front-end stage precisely because the recogniser was trained on cleaner data than it receives in deployment.
- **Hearing aids and cochlear implants.** Users consistently report that separating speech from background noise is their primary difficulty. Here the processing must also run within a few milliseconds of latency on a battery-powered device.
- **Forensic and archival audio.** Recovering intelligible speech from degraded surveillance recordings or historical archives is often the entire objective.

In each case the goal is not simply to reduce the measured noise power. It is to improve **intelligibility**, and these two objectives can diverge. An aggressive suppressor can raise signal-to-noise ratio substantially while destroying the spectral detail that makes consonants distinguishable.

1.3 Why the problem is hard

Three properties of speech make denoising genuinely difficult rather than a routine filtering exercise.

Speech is non-stationary. Its spectral content changes on the order of tens of milliseconds as the vocal tract reconfigures between phonemes. A single Fourier transform taken over an entire utterance averages all of these configurations together, and any filter derived from that averaged spectrum will be wrong for every individual instant. This rules out the naive approach of transforming the whole signal, thresholding the spectrum, and transforming back — an approach that works convincingly on a stationary test signal composed of a few sinusoids and fails on real speech.

Signal and noise overlap in frequency. Classical filter design assumes the bands of interest are separable. Broadband noise occupies the same frequency range as speech, so no fixed low-pass, high-pass or band-stop filter can separate them. The separation has to be made adaptively, per frequency bin and per time frame, on the basis of an estimate of which bins currently carry speech energy.

Suppression introduces its own artefacts. Methods that estimate a noise spectrum and remove it are subject to estimation error. Where that error causes isolated spectral peaks to survive in otherwise silent bins, the reconstructed signal contains short-lived tonal bursts at randomly varying frequencies. This artefact, known as **musical noise**, is often judged more objectionable by listeners than the original broadband noise, even though the measured SNR has improved. Characterising and mitigating this effect is a central concern of this project.

1.4 Scope and approach

This project restricts itself to **classical, non-learned signal processing methods**. Deep learning approaches now dominate published benchmarks, but classical methods remain the appropriate subject for this study for several reasons: they are analytically derivable and therefore interpretable, they require no training data, they operate at low latency on constrained hardware, and — most relevant to a DSP course — their behaviour follows directly from the transform theory being studied.

Three methods are implemented and compared: spectral subtraction, Wiener filtering, and wavelet thresholding. The first two operate in the short-time Fourier domain; the third provides a deliberate contrast by operating in a different basis with different time-frequency localisation properties.

2. Objectives

The project is organised around the following objectives.

O1. Implement a correct short-time Fourier analysis–synthesis framework. Construct framing, windowing, transform, inverse transform and overlap-add stages such that analysis followed by synthesis with no intermediate modification reconstructs the input signal to within numerical precision. This perfect-reconstruction property is the foundation on which every subsequent result depends and will be verified explicitly before any denoising is attempted.

O2. Implement three denoising methods from first principles. Spectral subtraction, STFT-domain Wiener filtering, and wavelet soft thresholding, each written directly from its governing equations using primitive numerical operations rather than invoking an existing denoising library.

O3. Construct a controlled evaluation dataset with exact ground truth. Mix clean speech with noise at precisely specified signal-to-noise ratios, covering both stationary (white Gaussian) and non-stationary (babble) noise, across an input SNR range of -5 to $+15$ dB.

O4. Evaluate using both signal-fidelity and perceptual metrics. Report segmental SNR, PESQ and STOI, and examine the extent to which they agree. A specific sub-objective is to demonstrate cases where a method improves a fidelity metric while degrading a perceptual one.

O5. Characterise the failure modes of each method. Identify and visually demonstrate musical noise in spectral subtraction, quantify the improvement obtained through over-subtraction and spectral flooring, and compare transient preservation between the Fourier-based and wavelet-based approaches.

O6. Produce a reproducible software deliverable. A documented, modular, seeded codebase that regenerates every figure and table in the final report from a single entry point.

3. Methodology

3.1 Short-time Fourier transform framework

All Fourier-domain processing operates on overlapping frames rather than the complete signal. The signal is segmented into frames of 32 ms with 50% overlap and each frame is multiplied by a Hann window $w[n]$ before transformation:

$$X(m, k) = \sum_{n=0}^{N-1} x[n + mH] w[n] e^{-j2\pi kn/N}$$

where m indexes the frame, k the frequency bin, H the hop size and N the frame length.

The 32 ms frame length is a deliberate compromise. Speech is approximately stationary over intervals of 20–40 ms, so frames must be short enough to fall inside that window; but frequency resolution is inversely proportional to frame length, so frames that are too short cannot resolve the harmonic structure of voiced speech. The Hann window is chosen for its low spectral leakage, and the 50% overlap with a Hann window satisfies the constant overlap-add (COLA) condition, guaranteeing that the windowed frames sum back to unity and reconstruction is exact.

Phase handling. All three Fourier-domain methods modify only the magnitude $|X(m, k)|$ and reuse the noisy phase $\angle X(m, k)$ for reconstruction. Phase estimation is a substantially harder problem, and it is well established that the human auditory system is comparatively insensitive to phase distortion in this context. The reconstructed frame is therefore

$$\hat{S}(m, k) = G(m, k) |X(m, k)| e^{j\angle X(m, k)}$$

where $G(m, k)$ is the gain function defining the method. The methods below differ only in how G is computed.

3.2 Method 1 — Spectral subtraction

Following Boll (1979), the noise magnitude spectrum is estimated by averaging over frames identified as speech-absent, typically a leading silence region. The estimate is then subtracted from each frame:

$$|\hat{S}(m, k)| = \max\left(|X(m, k)| - \alpha |\hat{D}(k)|, \beta |X(m, k)|\right)$$

The over-subtraction factor $\alpha > 1$ removes more than the estimated noise, trading speech distortion for reduced residual noise. The spectral floor β prevents bins from being driven to zero, which is the mechanism by which musical noise arises. Both parameters are exposed as configurable inputs so that the $\alpha = 1, \beta = 0$ case can be run to demonstrate the artefact, and the effect of raising each can then be shown.

3.3 Method 2 — Wiener filtering

The Wiener filter is the linear estimator minimising the mean squared error $\mathbb{E}[(s - \hat{s})^2]$. In the STFT domain its gain is

$$G(m, k) = \frac{\xi(m, k)}{1 + \xi(m, k)}, \quad \xi(m, k) = \frac{\mathbb{E}[|S(m, k)|^2]}{\mathbb{E}[|D(m, k)|^2]}$$

where ξ is the a priori SNR, the ratio of clean speech power to noise power in that bin. Since the clean speech power is unknown, ξ is estimated using the decision-directed approach of Ephraim and Malah:

$$\hat{\xi}(m, k) = \eta \frac{|\hat{S}(m-1, k)|^2}{|\hat{D}(k)|^2} + (1 - \eta) \max(\gamma(m, k) - 1, 0)$$

which forms a weighted combination of the previous frame’s estimated speech power and the current frame’s instantaneous (a posteriori) SNR γ . The smoothing parameter $\eta \in [0, 1]$ directly controls the trade-off between artefact suppression and transient responsiveness, and will be swept experimentally.

3.4 Method 3 — Wavelet thresholding

As a non-Fourier contrast, the discrete wavelet transform decomposes the signal into coefficients localised in both time and scale, giving better time resolution at high frequencies than the fixed-resolution STFT. Under the assumption that speech energy concentrates in a few large coefficients while white noise spreads evenly across many small ones, denoising reduces to thresholding.

Donoho’s soft thresholding shrinks each detail coefficient w toward zero,

$$T_\lambda(w) = \text{sgn}(w) \max(|w| - \lambda, 0)$$

using the universal threshold

$$\lambda = \sigma \sqrt{2 \ln N}$$

where the noise standard deviation σ is estimated robustly from the median absolute deviation of the finest-scale detail coefficients w_1 :

$$\hat{\sigma} = \frac{\text{median}(|w_1|)}{0.6745}$$

Wavelet family and decomposition depth are treated as parameters to be tuned. The expected advantage of this method is in the preservation of transients such as plosive consonants, which the Fourier methods tend to smear.

3.5 Dataset and noise conditions

Clean speech is corrupted with noise programmatically so that ground truth is known exactly. A mixing routine scales the noise signal to achieve a specified target SNR:

$$\text{SNR}_{\text{dB}} = 10 \log_{10} \frac{P_s}{P_d}, \quad P_s = \frac{1}{N} \sum_n s[n]^2$$

so that achieving a target SNR requires scaling the noise by

$$c = \sqrt{\frac{P_s}{P_d \cdot 10^{\text{SNR}_{\text{dB}}/10}}}$$

Two noise types are used: white Gaussian noise as the stationary baseline, and babble noise as a non-stationary and spectrally speech-like case that is considerably harder. Each is applied at input SNRs of -5 , 0 , 5 , 10 and 15 dB. The NOIZEUS corpus provides the reference speech material, with a synthetic fallback signal so that the pipeline can be executed without external data.

3.6 Evaluation metrics

Segmental SNR is computed per frame and averaged, rather than globally:

$$\text{segSNR} = \frac{1}{M} \sum_{m=0}^{M-1} 10 \log_{10} \frac{\sum_{n \in \mathcal{F}_m} s[n]^2}{\sum_{n \in \mathcal{F}_m} (s[n] - \hat{s}[n])^2}$$

where $\mathcal{F}_m$ is the set of sample indices in frame m . Because the logarithm is taken inside the average rather than outside, every frame contributes equally. Global SNR, by contrast, is dominated by high-energy frames and therefore conceals what happens during low-level speech, which is precisely where intelligibility is most at risk. Frames falling below an energy threshold are excluded to prevent silent regions from producing unbounded negative values.

PESQ (ITU-T P.862) and **STOI** provide perceptually motivated measures. These are included because SNR-type metrics are known to correlate poorly with listener judgement — musical noise, in particular, can accompany a rise in SNR and a fall in perceived quality. Demonstrating this divergence is an explicit objective (O4).

Spectrographic comparison. Before-and-after spectrograms are produced for representative cases, since musical noise is far more legible as a visual pattern of isolated time-frequency spikes than as any single number.

3.7 Implementation and tools

Python 3 with NumPy and SciPy for the transforms, PyWavelets for the wavelet decomposition, SoundFile for audio I/O, Matplotlib for figures, and the `pesq` and `pystoi` packages for perceptual metrics. The codebase is organised into separate modules for the STFT framework, noise mixing, each denoising method, metrics, experiment orchestration and plotting, driven from a single entry point. Random seeds are fixed so that every reported result is reproducible.

3.8 Work plan

Phase	Task
1	Literature review and method selection
2	STFT framework and perfect-reconstruction verification

Phase	Task
3	Noise mixing and metric implementation
4	Spectral subtraction
5	Wiener filtering
6	Wavelet thresholding
7	Full experimental sweep and analysis
8	Final report

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